

Analysis of Gut Microbiome Dynamics Following Supplementation with Orange and Raspberry Fruits

Gonalo Rego Gomes¹, Andreia Salvador^{1,2}, and Clarisse Nobre^{1,2}

¹ Universidade do Minho, Braga, Portugal

² CEB – Centre of Biological Engineering, Universidade do Minho, Braga, Portugal
goncalogomes999@gmail.com, {clarisse.nobre,
andreia.salvador}@ceb.uminho.pt

Abstract. Multiple diseases related to metabolic problems have been linked to alterations in the gut microbiota, a state known as dysbiosis. Meanwhile, polyphenols have emerged as potential prebiotic modulators capable of restoring intestinal well being, the analysing these complex interactions requires standardized computational approaches. The main objective of this project is to evaluate the prebiotic effects of fruit polyphenols by analysing the microbiota, polyphenol profiles and the production of short-chain fatty acids (SCFAs) by developing an automated bioinformatics pipeline for the post processing of 16S rRNA sequencing data. The final outcome of this project is a reproducible workflow that identifies microbial patterns and functional indicators across multiple donors, contributing to a better understading of personalized dietary interventions.

Keywords: Gut Microbiome · Polyphenols · Prebiotics · 16S rRNA Sequencing · Bioinformatics Workflow · Post-processing Automation

1 Objectives

The main objective of this project is to evaluate the prebiotic effects of fruit polyphenols by analysing the microbiota, polyphenol profiles and the production of short-chain fatty acids. Accordingly, this project aims to achieve this through a bioinformatics-centered approach by carrying out the following tasks:

- To review the state of the art regarding the profiles of healthy and pathological gut microbiota and the bioinformatics tools used for their analysis;
- Conduct a taxonomic analysis of microbial communities collected via Illumina 16S rRNA sequencing;
- Assess the variation in microbial communities based on dietary interventions;
- Relate microbial diversity to healthy and impaired microbial profiles;
- Develop a bioinformatics script to automate the analytical workflow, ensuring efficiency and reproducibility via a GitHub repository.

2 State of the Art

2.1 Gut Microbiome and Dysbiosis

Metabolic diseases such as obesity and type 2 diabetes have increased significantly in recent decades. These conditions are often associated with alterations in gut microbiota composition, which plays a key role in digestion, energy metabolism and immune regulation. [13]. A healthy microbiota is typically dominated by phyla such as Firmicutes, Bacteroidetes, Actinobacteria and Proteobacteria, maintaining functional stability and contributing to intestinal homeostasis. [23]. Dysbiosis is characterised by reduced microbial diversity and an imbalance between beneficial and potentially pathogenic species. [6]. Dysbiosis is often associated with a decrease in beneficial bacteria, including *Akkermansia muciniphila* and *Bifidobacterium*, and an increase in opportunistic microorganisms, particularly from the phylum Proteobacteria. [?, 1, 18]. In this context, the modulation of the gut microbiota has been investigated on a large scale as a tool for restoring gut eubiosis. Among the various approaches to achieve this, one may resort to dietary interventions, lifestyle changes, and the administration of probiotics, specifically strains belonging to the genera *Lactobacillus* and *Bifidobacterium* [10].

2.2 Emergence of polyphenols as prebiotics

Recently, the scientific community has shown a strong interest in prebiotics. Prebiotics are substrates that are selectively utilised by the host's microorganisms and confer health benefits [7]. Examples of these compounds include non-digestible carbohydrates, such as fructooligosaccharides and galactooligosaccharides, which are resistant to digestion in the small intestine and undergo fermentation by the microbiota in the colon, promoting the growth of beneficial bacteria, such as bifidobacteria [5, 8]. In some more severe clinical cases, more complex therapeutic treatments are chosen, such as faecal microbiota transplantation, to attempt to restore the balance of the gut microbiota [3]. In recent years, there has been a growing interest in the role of polyphenols as modulators of the gut microbiota. Polyphenols constitute a heterogeneous group of bioactive compounds and are found in various plant-based foods, such as fruits, vegetables, whole grains, tea and wine. Among the main classes, flavonoids, phenolic acids, stilbenes and tannins stand out; these compounds have been investigated for their antioxidant, anti-inflammatory and potentially cardioprotective properties [20]. One of the most notable properties of polyphenols is that they are poorly absorbed in the small intestine, meaning they reach the colon virtually intact, where they are metabolised by the gut microbiota. During this phase, the bacteria convert the polyphenols into low-molecular-weight phenolic metabolites, which may exhibit high biological activity. Thus, a bidirectional interaction is established, as whilst the microbiota transforms the polyphenols, these in turn influence the composition and activity of the microbial communities [4].

2.3 Functional Interpretation and Metabolites

Polyphenols act as selective substrates, promoting the growth of beneficial taxa such as *Bifidobacterium* and *Lactobacillus*, whilst potentially limiting the proliferation of other potentially pathogenic bacteria. It has been observed that the consumption of products rich in these compounds leads to positive changes in gut health and an increase in the presence of microbiota beneficial to the host's metabolism [15]. Other studies also suggest that dietary polyphenols are linked to an increase in the abundance of beneficial bacteria such as *Faecalibacterium prausnitzii* and *Anaerostipes*, which are associated with improvements in metabolic and inflammatory parameters [19]. Among the metabolites produced by fermentation are short-chain fatty acids (SCFAs), such as acetate, propionate and butyrate. These compounds play a central role in maintaining gut health. Butyrate, in particular, is the main source of energy for colonocytes and contributes to maintaining the integrity of the intestinal barrier. In turn, propionate and acetate play roles in various systemic metabolic processes, such as the regulation of lipid and glucose metabolism [12]. To enable the investigation of interactions between dietary compounds and the gut microbiota, several studies employ in vitro experimental models that simulate the human digestive process, followed by colonic fermentation. These models will allow for the controlled assessment of the impact of different bioactive compounds on the composition of faecal microbiota under anaerobic conditions [16,21].

2.4 Bioinformatics tools

Microbiota characterisation is frequently carried out using next-generation sequencing (NGS) techniques, with 16S rRNA gene sequencing being one of the most popular methodologies. This approach allows the identification and quantification of the various groups of bacteria present in the sample, enabling the analysis of diverse microbial communities and the comparison of changes in their composition under various experimental conditions [9,11]. While the initial data processing is often performed by external sequencing providers, post-processing and downstream analysis are critical for biological interpretation. Platforms such as QIIME2 are commonly used to perform standardized workflows for these types of analysis [2], therefore, this workflow is designed to handle processed abundance tables and includes several key stages:

- **Diversity analysis:**
 - Alpha diversity (e.g., Shannon, Simpson indices) to assess within-sample richness and evenness;
 - Beta diversity (e.g., Bray-Curtis, UniFrac distances) to evaluate differences between samples through PCoA visualizations;
- Taxonomic profiling:
 - Determination of relative abundances of microbial taxa across different taxonomic levels;
 - Identification of dominant and differential taxa between experimental groups;

- Statistical analysis:
 - Multivariate methods such as PERMANOVA to assess group differences;
 - Differential abundance analysis (e.g., ANCOM) to identify significantly altered taxa;
- Data integration:
 - Correlation of microbiome composition with metabolic data, such as SCFA concentrations and polyphenol profiles, enabling a functional interpretation of microbial changes.

An important contribution of this project is the development of a dedicated bioinformatics script to automate this analytical pipeline. The script should include various post processing steps, from data input to the generation of visual outputs, ensuring a standardized and efficient analysis that can be applied to data from multiple donors. Both scripts and workflows will be maintained in a **GitHub repository** to promote reproducibility on this type of analysis. This bioinformatic framework will be applied in the present project to analyse microbiome data obtained from in vitro fermentation assays, enabling the identification of microbial patterns associated with different polyphenol-rich dietary interventions.

3 Problem Analysis

Despite the growing interest in diet-microbiota interactions, the specific mechanisms by which polyphenols modulate the microbiome remain poorly understood due to high inter-individual variability. [22]. The effects of polyphenols can vary significantly between individuals, due to the wide variation in the initial composition of the microbiota and also in the microbial metabolism of these compounds. Therefore, it is important to analyse changes in microbial composition and metabolite production in response to various dietary interventions in an integrated manner [17]. This project addresses this gap by implementing an automated bioinformatics workflow to evaluate the prebiotic effects of orange and raspberry supplements. By processing 16S rRNA data and correlating it with polyphenol and SCFA profiles, this study aims to provide a standardized approach to understanding dietary impacts on the gut microbiome.

4 Methodology

The methodology is structured as a reproducible bioinformatic pipeline composed of sequential analytical steps.

4.1 Data Input

The dataset will be provided by the company in a pre-processed format, including feature tables (ASVs/OTUs), taxonomic assignments, and associated metadata describing experimental conditions.

4.2 Bioinformatic Workflow

The analysis will be performed using QIIME2 and complementary tools in R (e.g., phyloseq, vegan), following a structured workflow:

1. Data Import and Validation

Import of feature tables, taxonomy, and metadata into the analysis environment, followed by consistency checks.

2. Diversity Analysis

- Calculation of alpha diversity metrics (Shannon, Simpson);
- Calculation of beta diversity distances (Bray-Curtis, UniFrac);
- Ordination analysis (PCoA) for visualization of sample clustering.

3. Taxonomic Profiling

- Determination of relative abundances;
- Identification of dominant taxa;
- Comparison of taxonomic composition across experimental conditions.

4. Statistical Analysis

- Application of PERMANOVA to assess group differences;
- Differential abundance analysis using ANCOM;
- Correction for multiple testing where applicable.

5. Data Integration

Integration of microbiome data with:

- Polyphenol profiles;
- SCFA concentrations;

to identify correlations between microbial taxa and metabolic outputs, enabling functional interpretation of microbiome changes.

6. Visualization

Generation of graphical outputs, including barplots, heatmaps, and ordination plots, to support data interpretation.

7. Workflow Automation

Development of a bioinformatics script (R/Python) to automate the full analysis pipeline, from data input to result generation, ensuring reproducibility and efficiency.

The overall bioinformatic workflow implemented in this project is illustrated in Figure 1.

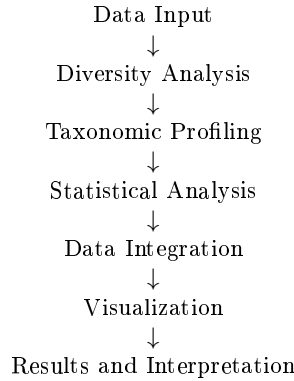


Fig. 1. Bioinformatic workflow for this project.

The workflow will be implemented in a reproducible environment and version-controlled using GitHub, ensuring transparency and reusability of the analysis.

5 Work Plans

The execution of this project is organized into three distinct operational phases, as defined by the course regulations. The timeline and task distribution are summarized in Table 1.

Phase 1: Problem Study and Preliminary Definition. This phase focuses on the literature review regarding healthy and pathological gut microbiota profiles [18]. It includes the preliminary analysis of the problem and the definition of the 16S rRNA sequencing methodologies to be used [9, 11]. The primary outcome is the delivery of an interim article (max. 6 pages).

Phase 2: Project Development and Results Analysis. This stage involves the taxonomic analysis of microbial communities under dietary interventions (orange and raspberry supplements) [17]. It culminates in a public presentation (max. 12 minutes) to report the main findings and incorporate feedback into the final results.

Phase 3: Final Reporting. The final phase consists of synthesizing the integrated results—microbiota composition, polyphenol profiles, and SCFA production [12]—into a comprehensive final article (max. 12 pages).

Table 1. Project development timeline based on the three mandatory phases.

Phase	Key Tasks	Outcome
Phase 1	State of the art, problem analysis, and methodology definition.	Interim Article
Phase 2	Data processing (QIIME2), statistical analysis, and presentation.	Public Presentation
Phase 3	Final integration of metabolic data and discussion.	Final Article

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